Two-mode Network Analysis of Culture Concepts

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Abstract

Kroeber and Kluckhohn's classic 1952 monograph *Culture: A Critical Review of Concepts and Definitions* (Kroeber and Kluckhohn, 1952) brought together over a hundred scholarly definitions of culture and indexed the key concepts they discuss. We build one network from that index to explore how concepts and definitions are connected. In the *definitions network*, concepts like **society** and **beliefs** are linked to 171 definitions, grouped into seven categories and credited to the author or authors that K&K index under each. Each definition is one quotation, identified by its author, year, page numbers and opening words. We present basic statistics about the network, visualize the data using correspondence analysis and spanning trees, and highlight the key concepts and definitions that anchor it. Data-driven types found in the concept profiles and in the wording of the definitions do not recover K&K's categories at better than chance.

1 Data and networks

We build one network from the Index B of Kroeber and Kluckhohn (1952), hereafter K&K. The *definitions network* links 470 concepts—concrete ideas like **society**, **customs**, and **beliefs**—to 171 definitions of culture, which K&K number as definitions. Each definition is credited to the author or authors K&K index under it, and falls into one of seven categories, which K&K label with the capital letters A through G. A Descriptive, B Historical, C Normative, D Psychological, E Structural, F Genetic, and G Incomplete.[1] Of the 171 definitions, 169 appear in K&K's own author index. The other two, C-I-18a and D-II-16, are attributions cited in the main text of the book that the index does not list. Boroch (2016) quotes two parts of four K&K definitions separately, lettered (a) and (b). A-3, E-5, F-II-4 and F-II-7. We keep each of these as two definitions, A-3(a) and A-3(b) and so on, which is why the eight variant definitions replace four. K&K print one index entry per definition, so each variant starts with its parent's concept links. For 29 of the 41 concept links copied in this way, we kept the link only on the variant whose quotation contains the concept's name; the other 12 remain on both variants. That assignment is our reading of the quotation text, not taken from anything directly printed in the K&K index. Concepts have consistent names; where a concept appears under different headings in K&K's source text, we treat it as a single concept. Each definition is one quotation, identified by its author, year, page numbers and opening words. Appendix A describes how each definition was matched to its quotation and how the a and b variants were built.

The 171 × 470 definitions network naturally breaks into 10 separate components, with the largest containing 626 nodes. Our analysis focuses on this largest component, which contains vir-

[1] K&K further divide three of the seven categories into numbered subcategories. Normative (C) into C-I and C-II, Psychological (D) into D-I through D-IV, and Genetic (F) into F-I through F-IV. The other four categories have no subcategories. Definitions that sit in a subcategory carry its numeral, as in C-I-18a or F-II-4 below.

Table 1: Summary statistics for the definitions network. Statistics describe the full network, before the restriction to the largest component's 2-core used in the analyses. Average nearest-neighbor degree (k_{nn}) is the average degree of a node's neighbors; degree correlation shows how a node's degree relates to the degree of its neighbors. Degree variances are sample variances (divisor $n-1$) over all nodes of each type, including isolates.

	Network
Edges	1201
Density	0.015
Degree correlation r	−0.076
Concepts ($N = 470$)	
Mean degree	2.56
Minimum degree	1
Maximum degree	36
Degree variance	16.34
Mean k_{nn}	10.18
Definitions ($N = 171$ definitions, 5 isolated)	
Mean degree	7.02
Minimum degree	0
Maximum degree	25
Degree variance	20.52
Mean k_{nn}	9.22

tually all the meaningful connections. We use maximum spanning trees across multiple projections to identify the strongest connections in different views of the network.

2 Overview of the network

Table 1 gives summary statistics for the definitions network. The first three rows show node and edge counts. The "degree" rows count connections of each node type; a concept's degree is how many definitions are linked to it in K&K's index, while a definition's degree is how many concepts it is linked to in the index. Concepts and definitions differ substantially in their average degree.

The degree correlation is −0.076, implying that concepts with many connections tend to link to definitions with few, and vice versa, though the association is weak.[2] The nearest-neighbor rows show the other side of this pattern. A concept's neighboring definitions contain 10.2 concepts on average, and a definition's neighboring concepts appear in 9.2 definitions on average, so concepts attach preferentially to heavily used definitions and definitions attach to widely shared concepts (Barabási and Albert, 1999)

Figure 1 shows the full-network degree distributions for the concept nodes (left) and the definition nodes (right). Most concepts are poorly connected; 79% appear in two or fewer definitions, while a small minority of concepts, such as **social**, **society**, **group**, **behavior** or **ideas**, are "super-

[2]Against a degree-preserving null that swaps memberships between random pairs of concepts until the mixing reaches a limit (1000 draws; seed 6128), the null mean is −0.033 with SD 0.022 ($z = -1.94$), and the observed value sits at the 4.1% point of the two-sided null distribution ($p = 0.041$). The negative association is therefore marginally stronger than degree structure alone would produce.

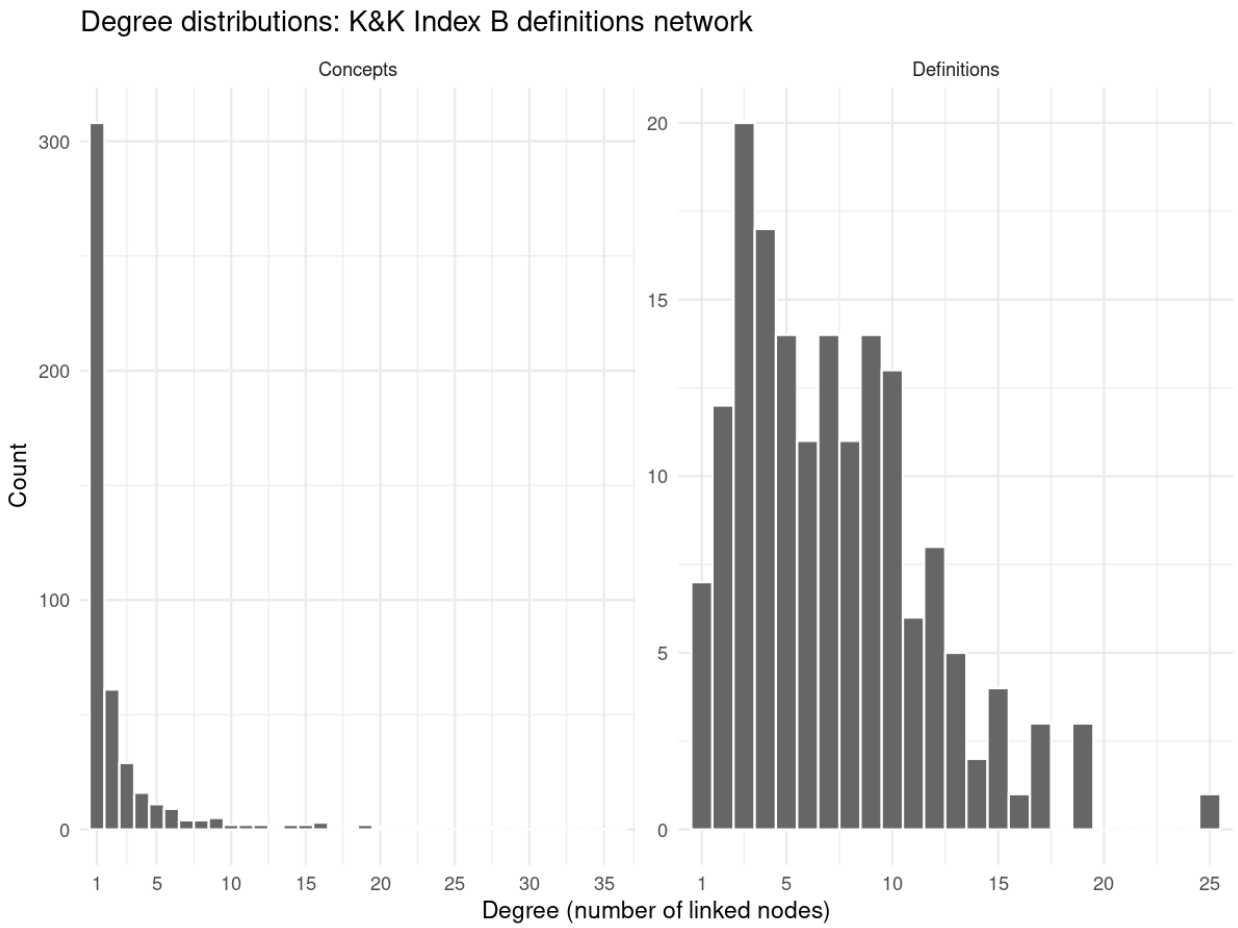


Figure 1: Degree distributions of the definitions network. On the left, sub-element (concept) nodes; on the right, definition nodes.

connectors" appearing in many definitions (see Table 2). This leads to the classic "long tailed" degree distribution for definitions observed in the left-hand side of Figure 1.[3]

Definitions are even more evenly connected, with the average definition covering 5.9 concepts (median 5) and a few covering many more, up to 15 (Table 2).[4] The degree distribution differs from that of the definitions accordingly, coming closer to a bell-shape with a well-behaved (representative) mean. Thus, few definitions are "lean" (relying on a few concepts) and even fewer are conceptually profligate, with most definitions relying on a moderate number of concepts. This means that while most definitions help themselves to a small set of ubiquitous concepts, they also draw from a wider set of concepts not necessarily shared across definitions.

[3]Over the full range of concept degrees, a discrete power law fits better than a truncated exponential (Vuong $z = 6.1$, $p < 0.001$; $\hat{\alpha} = 2.13$), suggesting a mechanism of "preferential attachment" of definitions to already popular concepts (Barabási and Albert, 1999). The advantage shrinks for degrees of 2 and up ($z = 1.9$, $p = 0.06$) and vanishes for degrees of 3 and up ($|z| < 1$), so the tail alone does not distinguish the two.

[4]Across the 171 definitions, the number of words in each correlates with the number of 2-core concepts the definition links to. Pearson $r = 0.55$, Spearman $\rho = 0.57$. Excluding the 22 definitions outside the 2-core gives $r = 0.60$ and $\rho = 0.61$ on 149 definitions. Word counts use the English translation for the four non-English quotations and exclude Boroch's ellipsis markers.

Table 2: Highest-degree nodes of the definitions network on the LCC 2-core. Concept degree counts the 2-core definitions that contain the concept; definition degree counts the 2-core concepts the definition contains. The last two columns rank definitions by degree per 10 words of quotation, using the English translation where one exists. Per-word rankings favor short quotations; across the 171 definitions the per-word value correlates −0.23 with word count. Ties at the cutoff are included.

Concepts, by degree		Definitions, by degree		Definitions, by degree per 10 words	
Concept	Degree	Definition	Degree	Definition	Per 10 words
SOCIAL	35	F-I-8 (Reuter, 1939)	15	A-3(a) (Dixon, 1928)	5.00
SOCIETY	31	F-II-8 (Taylor, 1948)	15	D-II-4 (Kluckhohn, 1942)	4.29
GROUP	27	A-8 (Hiller, 1933)	14	D-III-2 (Young, 1934)	4.29
BEHAVIOR	24	C-I-6 (Gillin and Gillin, 1942)	14	F-III-1 (Bain, 1942)	4.29
TOTAL	22	C-II-4 (Bidney, 1946)	13	A-1 (Tylor, 1871)	4.14
IDEAS	21	D-I-7 (Panunzio, 1939)	13	B-13 (Groves and Moore, 1940)	3.75
BELIEFS	19	E-3 (Ogburn and Nimkoff, 1940)	13	G-4 (Rouse, 1939)	3.75
INDIVIDUALS	19	F-I-19 (Murdock, 1949)	13	F-II-6 (Kluckhohn and Kelly, 1945)	3.64
SUM	19	F-I-3 (Folsom, 1931)	13	B-6 (Malinowski, 1931)	3.46
CUSTOMS	16	A-1 (Tylor, 1871)	12	A-3(b) (Dixon, 1928)	3.33
LEARNING	16	C-I-17 (Frank, 1948)	12	F-I-10 (Dodd 1941)	3.33
				F-I-15 (Carr, 1945)	3.33

Note: Degrees are computed on the LCC 2-core used in the analyses (149 definitions and 160 concepts). Table 1 and Figure 1 describe the full network instead. The 22 definitions outside the 2-core have degree 0 and a per-10-word value of 0. Per-10-word values are definition degree × 10 divided by the word count of the quotation. Ties at the cutoff are included, so the three lists have different lengths. F-I-10 is Dodd 1941 as Boroch attributes it, a 1941 committee mimeograph of four authors.

3 Maximum spanning trees

By construction, the two-mode definitions network has no ties between two concepts or between two definitions, so we analyze the implied concept-to-concept ties and the definitions-to-definitions connections via dual one-mode projections (Everett and Borgatti, 2013). Following Breiger (1974), two concepts are tied if they share at least one definition, with weight equal to the number they share. Likewise, two definitions are tied through the concepts they share. In matrix form, the projections are:

$$W_{\text{concept}} = AA^{\top}, \qquad W_{\text{def}} = A^{\top}A, \tag{1}$$

Where $(W_{\text{concept}})_{ij} = \sum_k a_{ik}a_{jk}$ counts the definitions that include both concepts i and j. Each projection is a weighted graph whose edges are the off-diagonal pairs with positive weight. We build projections from the 2-core of the network's largest component.

Projected graphs are typically dense and fully connected making it hard to detect their underlyinng structure. To do so, we extract the "backbone" (Neal, 2014) of each projection by building the maximum spanning tree (MST). The MST of each projection is found by computing the minimum spanning tree on negated weights, $-w$. Kruskal's (1956) method sorts edges from largest weight to smallest and adds each unless it would close a cycle, producing a tree that maximizes $\sum_{e\in T} w_e$. Because the 2-core is connected, each tree has exactly $|V| - 1$ edges.

Many projection weights tie, so a maximum spanning tree is not unique. We therefore build each tree by Kruskal's method with tied edges in a random order, repeat this 1000 times for each projection, and report each node's mean Freeman-normalized (Freeman, 1978) betweenness over those trees, together with its spread and the share of trees in which it ranks in the top ten. The

Table 3: Top 10 nodes by mean Freeman-normalized betweenness over 1000 random maximum spanning trees of each projection. Degree is the node's degree in the 2-core. Top-10 share is the percentage of trees in which the node ranks in the top ten. Concepts and definitions are ranked separately. Ties are broken at random in each tree, so the ranks below the first place are not robust. A-10 (10th, share 45%) and A-1 (11th, share 50%) differ by 1.1 points in mean betweenness.

Node type	Node	Author-year	Mean norm. betw. (%)	Top-10 share (%)	Degree
Concepts	GROUP		66.0	100	27
	SOCIAL		57.8	100	35
	SOCIETY		54.7	100	31
	BEHAVIOR		51.6	100	24
	CUSTOMS		36.9	100	16
	BELIEFS		25.7	97	19
	MEMBERS OF A SOCIETY		21.5	99	14
	HABITS		19.3	92	15
	IDEAS		15.8	60	21
	LEARNING		12.5	51	16
Definitions	C-I-6	Gillin and Gillin 1942	77.2	100	14
	A-16	Bidney 1947	35.3	77	11
	F-I-3	Folsom 1931	33.3	86	13
	C-II-4	Bidney 1946	29.1	77	13
	F-I-19	Murdock 1949	28.5	71	13
	C-II-3	Bidney 1942	26.9	71	10
	F-II-8	Taylor 1948	25.9	52	15
	A-8	Hiller 1933	24.1	77	14
	C-I-4	Klineberg 1935	21.8	70	8
	A-10	Linton 1936	21.3	45	11

Note: Each definition has an identifier in the format X-Y-Z, where X is a letter (A–G) for K&K's seven definition categories (Descriptive, Historical, Normative, Psychological, Structural, Genetic, Incomplete), Y is a roman numeral for the subcategory, and Z is the number of the definition within it. Variants such as A-3(a) and A-3(b) are the two quotations of one K&K definition, and they are counted as separate definitions (see the data section). The author–year column names the source of each definition's quotation, as Boroch attributes it; the same author and year label the node in Figure 3.

same procedure applies to the concept and definition projections. The tree shown in each figure is the ensemble member whose betweenness vector is closest to the mean (smallest squared distance). Node size in the figures shows the ensemble mean, so sizes need not match the betweenness of the tree shown, and node color shows the community structure detected on the tree shown using Newman's (2006) leading-eigenvector method.

The concept-level tree (Figure 2) shows how 160 concepts cluster in the concept projection and the definition-level tree (Figure 3) shows how 149 definitions cluster in the definitions projection. Table 3 lists the top 10 concepts and definitions by mean betweenness over the tree ensembles, with each node's top-ten share and 2-core degree.

Table 3 reports two kinds of measure for each node. Mean normalized betweenness is the share of the paths in a tree that run between other pairs of nodes and pass through the node, averaged over the 1000 trees and expressed as a percentage of the largest value a node could take in a tree, which is the center of a star. Betweenness indexes how much a node sits between otherwise separate parts of the projection. The top-10 share is the percentage of trees in which the node ranks in the top ten, so it measures how much the node's place depends on the arbitrary tie-breaking; a high mean with a low share is a tree-dependent result. Degree is a different quantity, the count of a

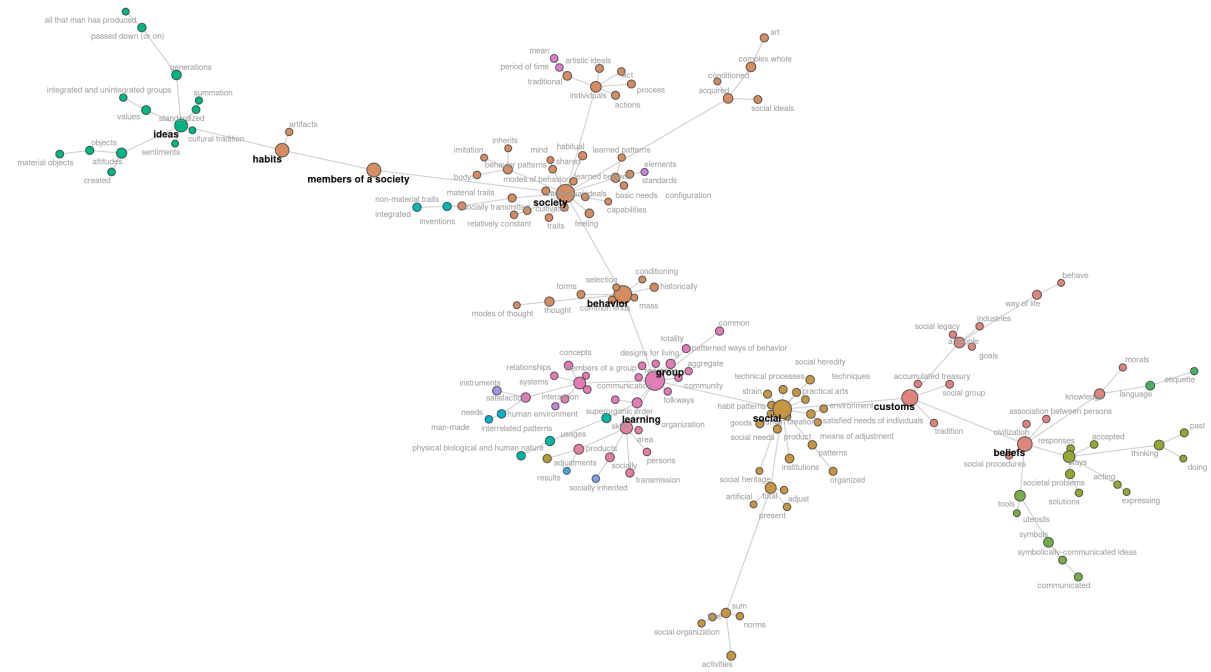


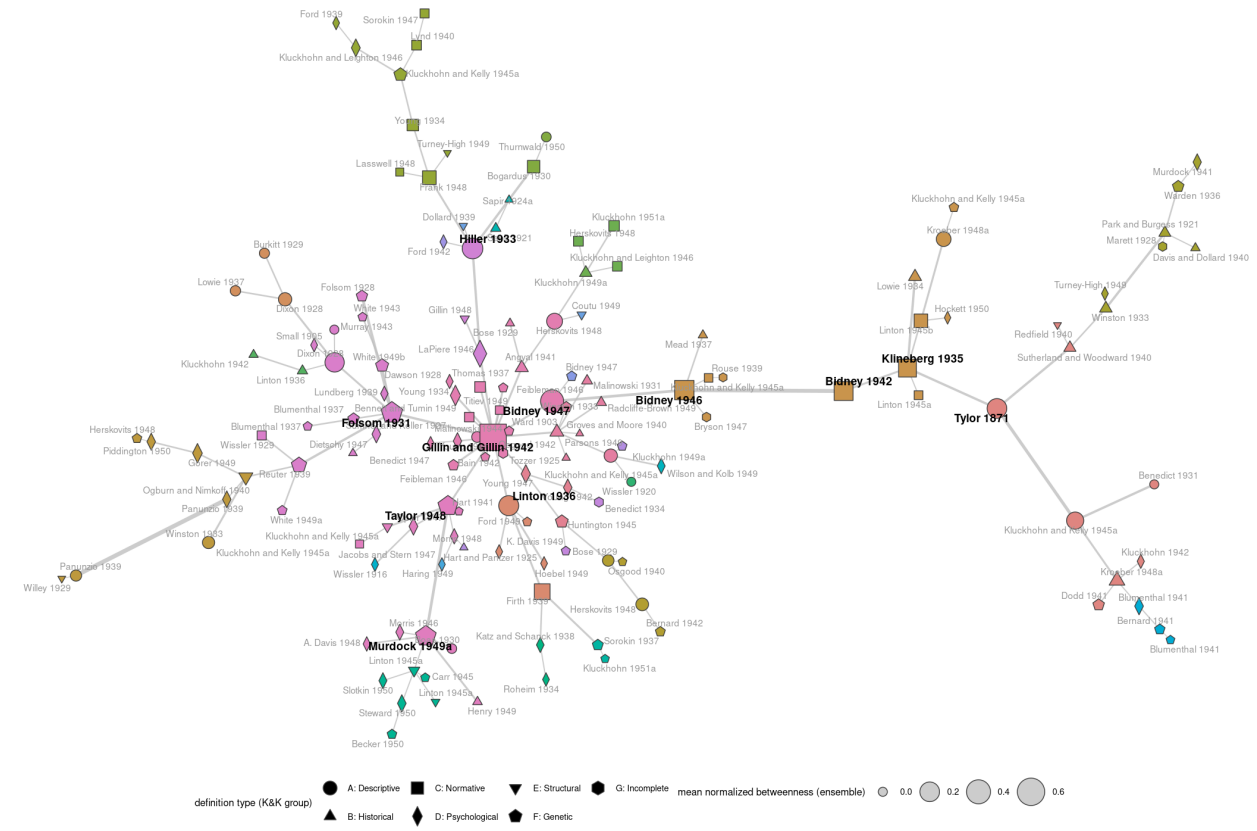
Figure 2: Maximum spanning tree of the definitions network, sub-element (concept) nodes. Color is the leading-eigenvector community on the tree shown; size is mean betweenness over the tree ensemble.

node's ties in the 2-core, and the two can diverge. **customs** has 16 ties and more than twice the betweenness of **ideas**, which has 21, while **learning** has 16 ties and only 12.5%. The same holds among definitions. C-I-6 has 14 ties but lies on 77% of the maximum, the largest share in the ensemble, which suggests it links regions of the tree that its degree alone would not reveal. The six highest-ranked concepts stay in the top ten in at least 97% of trees, so their places are robust, whereas A-10, in tenth place with a 45% share, is at the cutoff and could be replaced by A-1 in other draws. Substantively, the concepts at the top are the vocabulary that most often joins the definitions' regions in the tree, while the definitions with lower shares occupy positions that depend on which equally weighted tree is shown.

3.1 Concept tree

The concept tree is mostly a backbone of low-betweenness nodes. Averaged over the 1000 trees, 21 of the 160 concepts are leaves in every tree, and the median concept is a leaf in 76% of them. Concept degree (in the LCC 2-core) is positively associated with mean betweenness (Spearman $\rho = 0.71$), but not uniformly. The four highest-degree concepts, **social** (35), **society** (31), **group** (27), and **behavior** (24), are among the six highest in mean betweenness. Several other concepts with 15 or more 2-core definitions bridge little; *total* (22) has mean betweenness 7.3%, *sum* (19) 2.6%, and *transmission* (15) 0.9%. By contrast, **ideas** (21) reaches 15.8% and ranks ninth overall, so a degree similar to *total*'s can carry several times the betweenness, so **ideas** bridges more of the tree than *sum*, which has a similar degree (19 against 21).

In the concept tree (Figure 2), **group** is the hub of a pink region of *communication*, *organi-*


Maximum spanning tree, definitions network (code nodes), K&K Index B LCC 2-core
ties = shared sub-elements; labels = author and year of the quotation; shape = definition type; node size = mean normalized betweenness over the tree ensemble; black labels = top ten by mean betweenness
Hiller 1933
Folsom 1931
Bidney 1947
Gillin and Gillin 1942
Linton 1936
Taylor 1948
Murdock 1949a
Bidney 1946
Bidney 1942
Kluckhohn and Kelly 1945a
Kluckhohn and Leighton 1946
Kluckhohn and Leighton 1946
Kluckhohn 1949a
Kluckhohn 1951a
Herskovits 1948
Kluckhohn 1942
Kluckhohn 1949a
Kluckhohn and Kelly 1945a
Kluckhohn and Kelly 1945a
Kluckhohn and Kelly 1945a
Kluckhohn 1951a
Kluckhohn 1942
Kluckhohn and Kelly 1945a
Kluckhohn and Kelly 1945a
Kluckhohn 1942
Kluckhohn and Kelly 1945a
Kluckhohn and Kelly 1945a
Kluckhohn and Kelly 1945a

core only through **group**.

3.2 Definitions tree

Of the ten definitions with the highest mean betweenness, seven carry a C (Normative) or F (Genetic) definition, and the other three, A-16, A-8, and A-10, are Descriptive. Bidney holds three of them (A-16, C-II-4, and C-II-3) and Linton one (A-10), while no Kroeber definition is in the top ten. A-17 ranks 25th.

The three definitions with the highest mean betweenness are the Gillin and Gillin (1942) quotation (C-I-6, 77%), the Bidney (1947) quotation (A-16, 35%), and the Folsom (1931) quotation (F-I-3, 33%). The Gillin quotation is the largest node in the figure. Gillin and Gillin (1942) heads the largest community (18 definitions, drawn from six of the seven groups), and Bidney (1947) sits in the same community. The Gillin quotation is wordy (64 words) but not among the wordiest (89th percentile of length among the 171 definitions), so its bridging role is not a simple function of its length but of the concepts it uses. Here is the definition in full, with the concepts it uses in bold:

> The **customs**, **traditions**, **attitudes**, **ideas**, and **symbols** which **govern social behavior** show a wide variety. Each **group**, each **society** has a set of **behavior patterns** (overt and covert) which are more or less **common** to the members, which are passed down from **generation** to **generation**, and **taught** to the **children**, and which are constantly liable to **change**. These **common** patterns we call the culture...

The two trees' betweenness rankings are each their own structural fact rather than echoes of one another. A concept's mean betweenness is only weakly related to the mean betweenness of the definitions that contain it (Pearson $r = 0.13$; Spearman $\rho = 0.30$), and the reverse relation is weaker still ($r = 0.08$; $\rho = 0.10$). Concepts gain concept-tree betweenness by bridging concept regions, not by riding on definitions with high definition-tree betweenness: **group** owes its top concept-tree rank to its position among concepts, not to its definition partners. The converse also appears. A-10 (Linton 1936) ranks tenth in definition betweenness with only **society** among the six highest in concept betweenness, so a definition can bridge definitions without containing any bridge concept. The apparent association between the top concepts and the top definition is largely one hub; five of the six concepts with the highest concept betweenness (**group**, **social**, **society**, **behavior**, and **customs**) link to C-I-6, which has the highest mean betweenness among the definitions of each of them. The sixth, **beliefs**, does not link to it; its highest-betweenness definition is F-I-3.

C-I-6 has the highest mean betweenness in the definition tree (77%), and its position is not a simple function of its length. At 64 words it sits at the 89th percentile of quotation length among the 171 definitions, well short of the longest quotation (142 words). It links to 14 concepts, and the mean concept betweenness of those concepts sits at the 89th percentile of definitions. Its degree is not the highest. F-I-8 and F-II-8 each link to 15 concepts. But a high-betweenness vocabulary is not sufficient; 93 other definitions share at least one of its five highest-betweenness concepts, and their mean betweenness is 4.7% against C-I-6's 77.2%. What is distinctive about C-I-6 is where it sits in the definition tree: it bridges definitions, and it does so with a concept repertoire that many other definitions share.

Bidney's (1947) is in many ways more impressive. The Bidney definition (A-16) ranks second in mean definition betweenness (35%), but it is short (only 34 words). The links of A-16 are a mix. Of its 11 concepts in the 2-core, three are among the six highest in concept betweenness (**social**, **behavior**, and **society**), and the mean concept betweenness of its linked concepts sits at the 68th

percentile. A-16 has one further link outside the 2-core. Its position in the tree in Figure 3 is what gives A-16 high betweenness, since removing it leaves components of 116 definitions, 31 definitions, and one definition. Here is the definition in full, with its linked concepts in bold. Eleven of the twelve concepts linked to A-16 appear verbatim in the text, among them **acquired**, **behavior**, **society** and **social**. The twelfth, **techniques**, appears in the singular as *technique*:

> [. . .] functionally and secondarily, culture refers to the **acquired** forms of technique, **behavior**, **feeling** and **thought** of **individuals** within **society** and to the **social institutions** in which they **cooperate** for the attainment of **common ends**.

Two other Bidney definitions, C-II-4 and C-II-3, share a community that contains neither C-I-6 nor A-16. First, we have C-II-4 (Bidney, 1946), the fourth highest mean definition betweenness (29%). It is a "Normative" definition according to K&K, and it shares its community with C-II-3 (Bidney, 1942) but not with C-I-6 (Gillin and Gillin, 1942) or A-16 (Bidney, 1947). Its quotation is 40 words long, and it links to 13 concepts, among them **behavior** (fourth highest concept betweenness) and **society** (third):

> An integral or holistic concept of culture comprises the **acquired** or **cultivated behavior**, **feeling**, and **thought** of **individuals** within a **society** as well as the **patterns** or **forms** of intellectual, social, and artistic ideals which human societies have professed **historically**.

Finally, we have C-II-3 (Bidney, 1942), the sixth highest mean definition betweenness (27%). It is also a "Normative" definition according to K&K. Its quotation is 40 words long, and it links to 10 concepts in the 2-core, among them **society** (third), **behavior** (fourth), and **members of a society** (seventh):

> A culture consists of the **acquired** or **cultivated behavior** and **thought** of **individuals** within a **society**, as well as of the intellectual, artistic, and social ideals which the members of the society profess and to which they strive to conform.

Folsom (1931) (F-I-3), the third highest in mean definition betweenness (33%), is a "Genetic" definition that reaches the top ten in 86% of trees. Its 75 words put it at about the 95th percentile of length among the 171 definitions.

> Culture is not any part of man or his inborn equipment. It is the sum total of **all that man has produced**: **tools**, **symbols**, most organizations, **common activities**, **attitudes**, and **beliefs**. It includes both **physical products** and **immaterial products**. It is everything of a **relatively permanent** character that we call **artificial**, everything which is **passed down** from one generation to the next rather than **acquired** by each generation for itself: it is, in short, **civilization**.

F-I-3 links to 13 concepts in the 2-core (16 in the full network). The strongest link of F-I-3 is to **beliefs**, which is among the six concepts with the highest mean concept betweenness. The other concepts of F-I-3 range from **tools**, **attitudes** and **symbols** to **generations**, **common** and **sum**, several of which have low concept betweenness. Unlike the Gillin community, which draws on six of the seven groups, the definition tree in Figure 3 places F-I-3 in a community of 15 definitions, eight of them Genetic, so it sits mostly with its own group. The high betweenness of F-I-3 therefore does not come from bridging groups in the same way; the tree-level evidence points to its link to

beliefs and its breadth of concepts, but we have not decomposed its betweenness path by path, and its tree neighbors should be checked on the figure.

Murdock (1949) (F-I-19) has the fifth highest mean definition betweenness (29%). Its 70 words link to 13 concepts in the 2-core, three of which are among the four concepts with the highest concept betweenness (**group**, **behavior**, and **society**), and it has six tree neighbors, among them Taylor (1948) (F-II-8) and E-5(b). Its quotation frames culture as the product of a process, the "interaction of learning and society":

> The **interaction** of **learning** and **society** thus produces in every human **group** a **body** of **socially transmitted adaptive behavior** which appears super–individual because it is **shared**, because it is **perpetuated** beyond the individual life span, and because its quantity and quality so vastly exceeds the capacity of any single person to achieve by his own unaided effort. The term culture is applied to such **systems** of **acquired** and transmitted **behavior**.

Kroeber's A-17 (65 words; pp. 8–9) ranks 25th by mean betweenness (5.7%) and links to 10 concepts in the 2-core. Only one of them, **behavior**, is among the four highest in concept betweenness. The others include **habits** (19.3%), **ideas** (15.8%), and **learning** (12.5%), while **mass** has no betweenness in any tree and **techniques**, **totality**, and **transmission** are below 1.2%. Its tree neighbors are C-I-8 and F-II-6. Its quotation lists the components of culture and then frames it as a total product:

> [. . .] the **mass** of learned and transmitted **motor reactions**, **habits**, **techniques**, **ideas**, and **values** – and the **behavior** they induce – is what constitutes culture. Culture is the special and exclusive product of men, and is their distinctive quality in the cosmos [. . .]. Culture [. . .] is at one and the same time the **totality** of products of social men, and a tremendous **force** affecting all human beings, **socially** and **individually**.

Taylor (1948) (F-II-8), seventh in mean definition betweenness (26%), is the longest quotation in the analysis (142 words), and it reaches the top ten in only 52% of trees, so its place is less secure than most of the top ten.

> By [holistic] culture as a descriptive concept, I mean all those mental constructs or **ideas** which have been learned or **created** after birth by an individual. [. . .] The term idea includes such categories as **attitudes**, meanings, **sentiments**, feelings, **values**, goals, purposes, interests, **knowledge**, **beliefs**, **relationships**, associations, [but] not [. . .] Kluckhohn's and Kelly's factor of "designs". By [holistic] culture as an explanatory concept, I mean all those mental constructs which are used to understand, and to react to, the experiential world of internal and external stimuli. [. . .] Culture itself consists of **ideas**, not processes. By a culture, i.e., by culture as a partitive concept, I mean a **historically** derived system of culture **traits** which is a more or less separable and cohesive segment of the whole–that–is–culture and whose separate traits tend to be **shared** by all or by specially designated individuals of a **group** or "**society**".

It links to 15 concepts in the 2-core, and its strongest links are to **group** and **society**, the two most central concepts, together with **beliefs**, **ideas** and **learning**. Its community has 11 definitions, including the Murdock (1949) definition (F-I-19), which lists it as a tree neighbor, and definitions from Morris, Opler, Boas and Hart. Its betweenness thus comes from the concepts it shares with the top of the tree rather than from a single bridge.

Hiller (1933) (A-8), eighth in mean definition betweenness (24%), is a Descriptive definition of 58 words that reaches the top ten in 77% of trees.

> The **beliefs**, systems of thought, **practical arts**, manner of living, **customs**, traditions, and all **socially** regularized **ways** of **acting** are also called culture. So defined, culture includes all the **activities** which develop in the **association between persons** or which are learned from a **social group**, but excludes those specific forms of **behavior** which are predetermined by inherited nature.

The Hiller definition (A-8) links to 14 concepts in the 2-core, led by **social**, **customs**, **beliefs** and **learning**. The Hiller definition sits in a three-code community with LaPiere (1946) (D-II-5) and Gillin (1948) (E-7, a Structural definition and not the C-I-6 quotation), a small branch of the tree in Figure 3. The high betweenness of A-8 reflects that branch, since all three definitions contain **social** and **customs**, and Hiller's definition also contains **beliefs** and **learning**.

Klineberg (1935) (C-I-4), ninth in mean definition betweenness (22%), is a Normative definition of 36 words that reaches the top ten in 70% of trees.

> [Culture] applies to that whole "**way of life**" which is determined by the **social** environment. To paraphrase Tylor it includes all the **capabilities** and **habits acquired** by an individual as a member of a particular **society**.

With eight links in the 2-core, C-I-4 has one of the smaller degrees among the top ten, yet it links to the two most central concepts, **social** and **society**, and to **members of a society** and **habits**. C-I-4 shares a 13-code community with the two Bidney Normative definitions, C-II-4 and C-II-3, which is consistent with the separation of the Bidney definitions from C-I-6 and A-16.

Last but certainly not least, Tylor (1871) (A-1) ranks eleventh in mean definition betweenness (20%), just below A-10, and reaches the top ten in half of the trees. K&K describe Tylor's *Primitive Culture* (1871) as the work that established the anthropological meaning of "culture", and their index lists this quotation as A-1.

> Culture, or **civilization**, [...] is that **complex whole** which includes **knowledge**, belief, **art**, law, **morals**, custom, and any other **capabilities** and **habits acquired** by man as a member of **society**.

The 29-word Tylor quotation links to 12 concepts in the 2-core, the top three of which are **society**, **customs** and **beliefs**, with **members of a society** and **habits** next. The community of A-1 has eight definitions, mostly on the right of the figure, including Kluckhohn and Kelly (1945) (A-14), and it is separate from the Bidney and Gillin definitions. The Tylor definition connects to other definitions in the anthropological and sociological tradition, but by the time K&K wrote, it occupied a relatively peripheral position.

4 Data-driven topics: latent Dirichlet allocation

Latent Dirichlet allocation (LDA) (Blei et al., 2003) classifies definitions by their wording, with a soft weight on each topic. The LDA model treats each document as a mixture of topics and each topic as a distribution over words, so every definition carries a weight on every topic. We ask whether topics estimated from the wording of the definitions, without K&K's coding and without the concept links, recover K&K's seven categories A–G (Descriptive to Incomplete).

Table 4: The twelve LDA topics. The label we give each, its five highest-weight ensemble terms, and the number of definitions whose dominant topic it is (the topic with the largest weight).

Topic	Label	Top terms	Definitions
1	individual acquired behavior	behavior, individual, pattern, acquired, member	13
2	human products and results	human, product, result, activities, learned	13
3	transmitted group behavior	behavior, group, generation, pattern, learned	23
4	shared group living	group, men, common, relationship, living	10
5	biological needs and nature	mean, human, concept, biological, whole	9
6	acquired habits as complex wholes	habit, acquired, member, includes, idea	8
7	cultural order of phenomena	human, cultural, phenomena, order, animal	10
8	sum total of group life	sum, total, individual, group, mean	21
9	material tools and ideas	idea, material, includes, attitudes, tool	20
10	customs, beliefs and knowledge	custom, belief, people, tradition, way	13
11	way of life of a people	way, life, group, people, things	22
12	man-made environment	behavior, environment, part, material, made	9

The unit of analysis is the clause. Each quotation is split into sentences and then at semicolons, and clauses of fewer than three words are dropped, so that a definition listing several elements is not forced into one blend. Stopwords are removed, along with the frame words culture, society, social and man and a short list of grammatical words, and plural forms are merged into their singulars where both occur. A term is kept if it appears in at least two clauses. The model is fitted to 265 clauses from the 171 definitions, over a vocabulary of 340 terms. One definition, C-I-7 (Simmons 1942), keeps no clause after the term filter as the only substantive word it contains ("mores") is too rare in the corpus. It carries a uniform membership and drops out of every binary analysis.

We fix the number of topics at $k = 12$. This value was chosen for interpretability, not selected by the data, as held-out perplexity is flat for k from 8 to 30. The model is a Gibbs-sampled LDA with $\alpha = 0.1$ (Griffiths and Steyvers, 2004), fitted with the `topicmodels` package (Grün and Hornik, 2011) from ten seeds. Topic numbers are arbitrary across seeds, so each seed's topics are aligned to the first seed by Hungarian matching on the cosine between topic-term vectors, which leaves a mean matched cosine of 0.72. The stored membership of each definition is the mean over seeds of its clause-level weights, weighted by clause length.

Table 4 lists the twelve topics with the labels we give them from their leading terms, and Figure 4 shows the ten highest-weight terms of each topic, with bars drawn from the seed-mean topic-term probabilities. The labels are our reading of the top words and are heuristic. In a corpus this small, a topic can reflect one author's phrasing, so the labels describe the top terms and do not

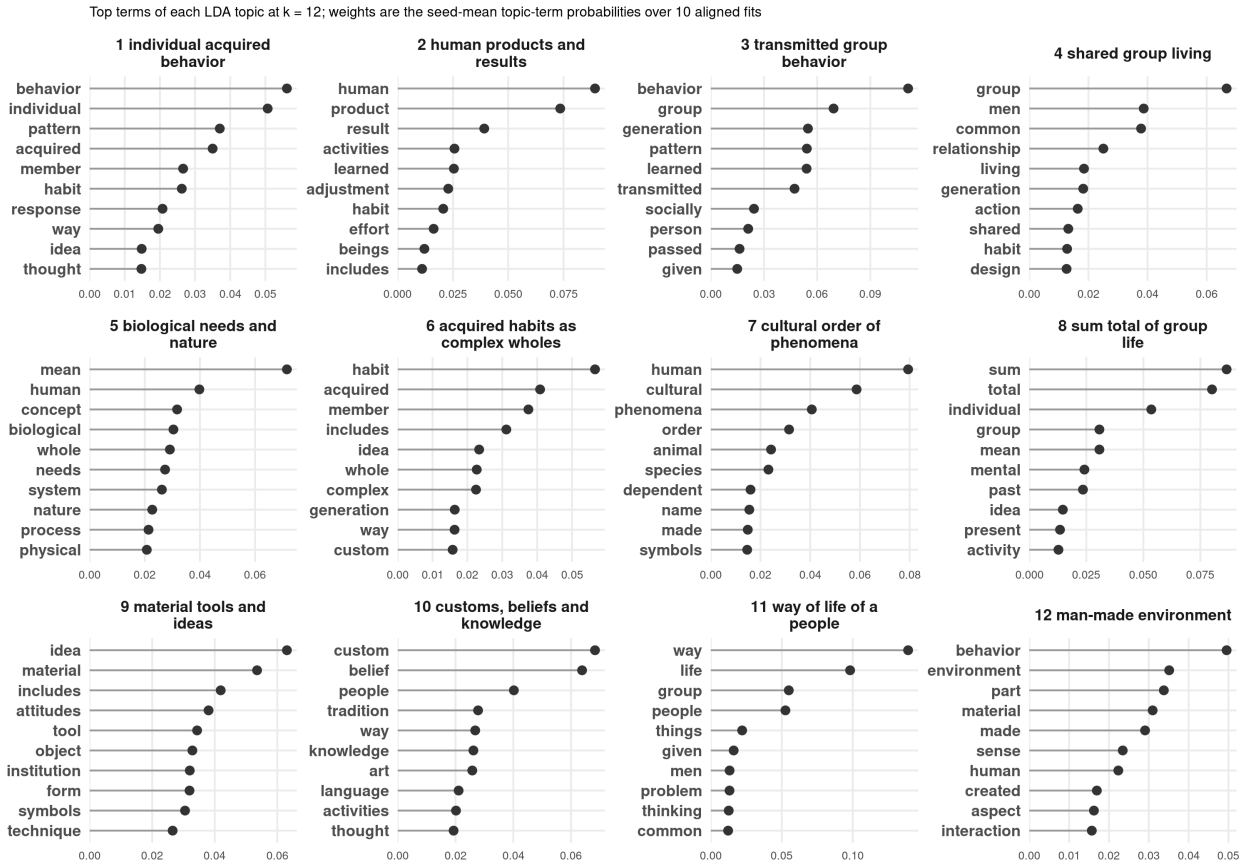


Figure 4: Top terms of each LDA topic at $k = 12$, labeled as in Table 4. Bars show the seed-mean topic-term probabilities over ten aligned fits.

name concepts. The topics differ in size. Dominant-topic counts run from 8 definitions (*acquired habits as complex wholes*) to 23 (*transmitted group behavior*), and the four largest topics together hold 86 of the 171 definitions. The heads of the largest topics are general words. *transmitted group behavior* is led by behavior, group and generation, and *way of life of a people* by way, life and group. Several smaller topics have more particular heads, such as *man-made environment* (environment, part, material, made) and *cultural order of phenomena* (human, cultural, phenomena, order).

Table 5 ranks the definitions within each topic by their weight on it and lists the five highest. The weight is the share of a definition's clauses assigned to the topic, so the ranking shows how strongly the wording of each definition leans toward the topic, not whether the definition belongs to it. C-I-7 has no clause in the model and is left out. Several top entries are short quotations. F-I-1 has five words, and the weights of F-I-9 and D-I-9 rest on twelve words each, so these ranks should be read with the length of the quotation in view.

4.1 LDA versus K&K's categories

The dominant topic of each definition, cross-tabulated against its K&K category, is non-zero but imperfect. A chi-square test of independence on the 7 by 12 table gives $\chi^2 = 114.5$, with a simulated $p = 0.0006$ (10,000 replicates, seed 8153). Expected counts are below 5 in 78 of the 84 cells, so the chi-square distribution does not apply, and the p-value comes from the simulation of the null hypothesis of independence. We report the statistic without a strength measure, since measures

of strength are unreliable on a table this sparse. Nevertheless, some topics do line up with the K&K categories. For instance, seven of the 22 Descriptive (A) definitions have *customs, beliefs and knowledge* as their dominant topic, and six of the ten definitions dominated by *cultural order of phenomena* are Genetic (F). Other topics mix categories. *Transmitted group behavior*, the largest, is led by Psychological (D) definitions at 9 of 23, and *way of life of a people* is led by Normative (C) definitions at 12 of 22. Cells this small should not be read as group-topic correspondences on their own.

The K&K Category composition of each topic is the most informative pattern, though the counts are small. These counts sum the subgroup-by-topic flows in the Sankey diagram (Figure 5), which links each definition's K&K subgroup to its dominant topic. No K&K category puts more than 44% of its definitions in one topic. The Normative (C) category is the most concentrated, with 12 of its 27 definitions are dominated by *way of life of a people*. Descriptive (A) definitions lean toward *customs, beliefs and knowledge* (7 of 22), and Genetic (F) definitions make up most of the dominant definitions of *man-made environment* (5 of 9) and a large share of *material tools and ideas* (9 of 20). Psychological (D) definitions, the largest category with 39 definitions, supply the most definitions to *transmitted group behavior* (9 of its 23 definitions), but no more than 23% of D definitions fall in any one topic.

Figure 5 repeats the category comparison at the finer grain of K&K's own typology, the roman-numeral subgroups the classification prints within categories C (Normative), D (Psychological), and F (Genetic). Categories A (Descriptive), B (Historical), E (Structural), and G (Incomplete) number their definitions plainly and have no subgroup level. Each definition is assigned to its dominant topic (the topic with the largest weight), and each ribbon takes the color of its origin node's group. Node size is the number of definitions. The panel covers all 171 definitions carrying a quotation.

The subgroups spread across the topics to different degrees. A (Descriptive), B (Historical), F-I and D-I reach ten, ten, nine and nine of the twelve topics as dominant topics, and C-I reaches seven. The largest cells are C-I on *way of life of a people* (12 of 21 definitions), D-II on *transmitted group behavior* (8 of 16), and A on *customs, beliefs and knowledge* (7 of 22). B on *transmitted group behavior* (5 of 23) and F-I on *sum total of group life* (4 of 21) are also among the larger cells. No subgroup's definitions sit on a single topic. The smallest subgroups, D-IV (2 definitions) and D-III (4 definitions), reach two and four topics, and F-III (5 definitions) reaches three. These counts describe only where each definition's largest weight falls. A definition with close weights on two topics is placed on one of them.

The subgroup-level flows in Figure 5 add three patterns. C-I is the most concentrated subgroup, with 12 of its 21 definitions fall in *way of life of a people*. Topic 10, *customs, beliefs and knowledge*, draws on only three subgroups, with A contributing 7 of its 13 definitions, B 4 and D-II 2. The Genetic subgroups spread more widely. F-I reaches nine of the twelve topics, with no topic holding more than 19% of its definitions, and F-I and F-II both contribute to *man-made environment* and *material tools and ideas*. *Transmitted group behavior* is the broadest topic, drawing on ten subgroups, and D-II supplies 8 of its 16 definitions to it. Subgroups with fewer than ten definitions (C-II, D-III, D-IV, F-III, F-IV and G) are too small for their shares to support comparisons.

Two cautions qualify these patterns. First, several words head more than one topic: group, habit, human, idea and way each appear among the ten highest-weight terms of four topics, and behavior of three. The topics therefore differ more in their secondary terms than in their headline words, and the labels summarize weight profiles rather than separate concepts. Second, most definitions are mixtures. The median definition puts 0.34 of its weight on its largest topic (a uniform membership would be 0.08), so the largest topic describes each definition only in part. This diffuseness is consistent with K&K's categories being spread across topics rather than matched to

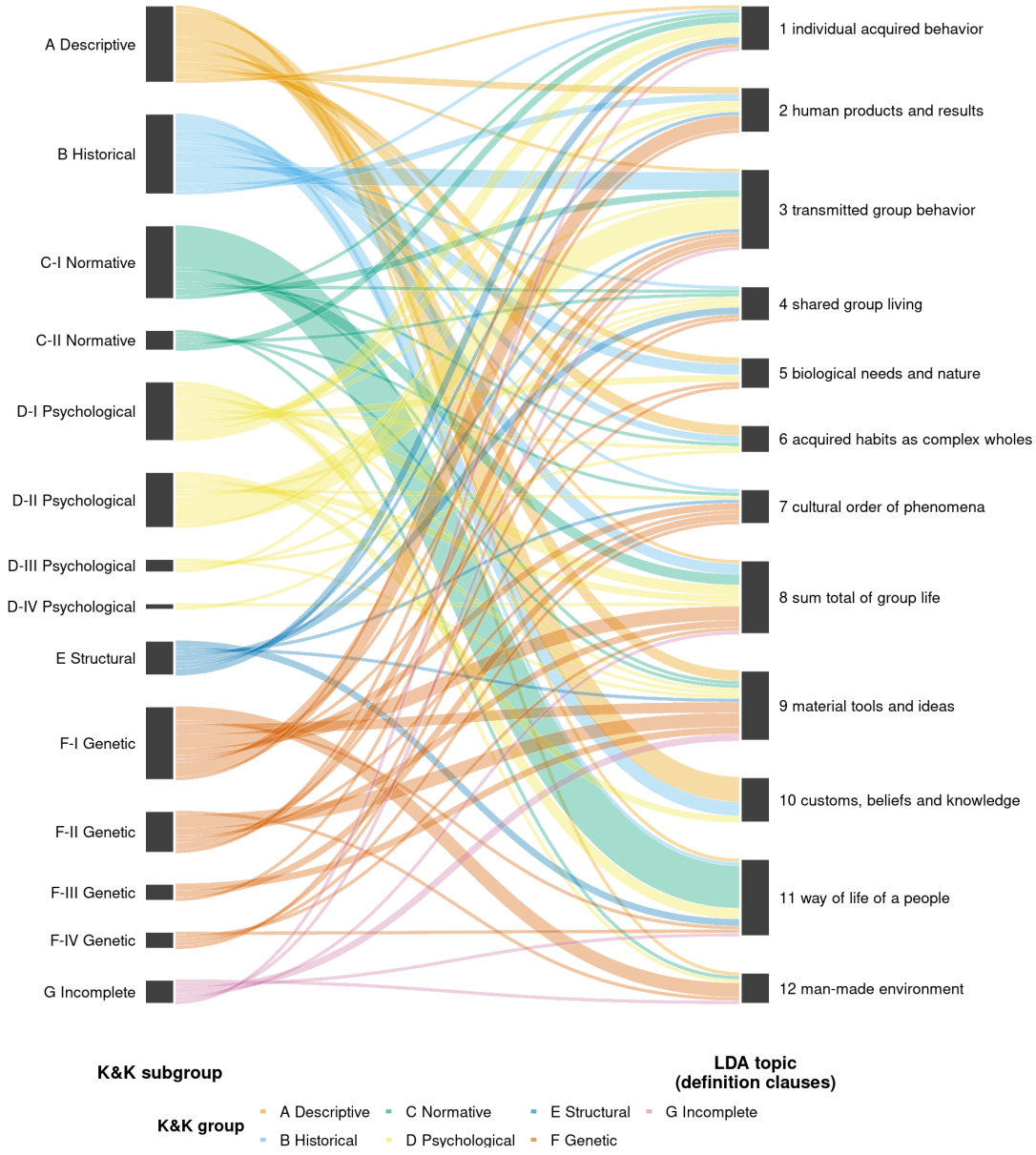


Figure 5: The 171 definitions carrying a text, by their own K&K subgroup (left) and dominant LDA topic (right). Ribbons take the color of their origin node's category.

them.

The topics recover themes in the wording of the definitions, but not K&K's taxonomy. One reading is that K&K's categories classify the emphasis of individual statements, and that a quotation of about thirty words cannot carry that emphasis. The other reading, which we find more plausible, is that the seven categories are editorial categories for organizing a bibliography rather than dimensions along which the wording varies. The analysis cannot separate these readings. The method's limits point the same way. The number of topics was fixed rather than selected, the quotations are the passages Boroch selected rather than the authors' full works, and four of them are rendered from translations that have not been checked against the page images.

5 Discussion

Table 5: The five definitions with the largest weight on each LDA topic. Each cell gives the definition's identifier with its weight in parentheses, and below it the work its quotation comes from. Boroch's appendix prints the reprint year 1931 for A-4, and the work is cited here as 1929. The four 1941 definitions (F-I-9, D-I-9, F-I-10 and F-I-11) are all one mimeographed work of the Committee on Conceptual Integration. Topic labels are those of Table 4.

Topic	Label	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
1	individual acquired behavior	C-II-4 (0.55) Bidney 1946	C-II-3 (0.53) Bidney 1942	E-1 (0.39) Willey 1929	B-21 (0.37) Henry 1949	A-10 (0.34) Linton 1936
2	human products and results	F-I-9 (0.73) Committee on Conceptual Integration 1941	D-I-9 (0.72) Committee on Conceptual Integration 1941	F-I-10 (0.60) Committee on Conceptual Integration 1941	F-I-1 (0.52) Groves 1928	D-II-13 (0.42) Hockett 1950
3	transmitted group behavior	D-II-6 (0.63) Benedict 1947	D-II-11 (0.56) Haring 1949	D-I-5 (0.55) Young 1934	D-II-14 (0.54) Steward 1950	B-14 (0.54) Angyal 1941
4	shared group living	B-5 (0.41) Bose 1929	E-6 (0.36) Kluckhohn and Kelly 1945	F-II-7(a) (0.32) Feibleman 1946	C-II-6 (0.30) Sorokin 1947	D-III-4 (0.29) Roberts 1951
5	biological needs and nature	A-11 (0.91) Panunzio 1939	D-I-7 (0.56) Panunzio 1939	B-9 (0.50) Linton 1936	D-I-16 (0.42) Gorer 1949	B-15 (0.33) Kluckhohn 1942
6	acquired habits as complex wholes	A-4 (0.56) Benedict 1929	B-8 (0.51) Lowie 1934	A-14 (0.44) Kluckhohn and Kelly 1945	A-1 (0.43) Tylor 1871	C-I-4 (0.38) Klineberg 1935
7	cultural order of phenomena	F-IV-2 (0.74) Ostwald 1915	C-II-1 (0.54) Carver 1935	F-III-4 (0.50) White 1949	F-I-11 (0.45) Committee on Conceptual Integration 1941	B-4 (0.41) Tozzer 1925
8	sum total of group life	A-10a (0.58) Lowie 1937	F-I-7 (0.50) Sorokin 1937	D-I-1 (0.45) Small 1905	B-1 (0.43) Park and Burgess 1921	B-20 (0.43) Kluckhohn 1949
9	material tools and ideas	F-III-2 (0.59) White 1943	C-I-18 (0.56) Titiev 1949	A-18 (0.48) Herskovits 1948	A-16 (0.46) Bidney 1947	C-II-2 (0.45) Thomas 1937
10	customs, beliefs and knowledge	A-2 (0.54) Wissler 1920	A-3(a) (0.53) Dixon 1928	A-3(b) (0.48) Dixon 1928	A-12 (0.48) Murray 1943	A-5 (0.44) Burkitt 1929
11	way of life of a people	C-I-13 (0.72) Kluckhohn and Leighton 1946	C-I-14 (0.49) Herskovits 1948	C-I-19 (0.48) Kluckhohn 1951	F-I-13 (0.45) Young 1942	C-I-5a (0.45) Lynd 1940
12	man-made environment	F-I-1a (0.62) Willey 1927	F-I-17 (0.58) Herskovits 1948	F-I-18 (0.54) Kluckhohn 1949	F-II-7(b) (0.49) Feibleman 1946	D-I-15 (0.47) Turney-High 1949

Appendix A Constructing the definition-level data

The analyses use one object, a table with a row for each of the 171 K&K definitions. Each row gives the definition's group and subgroup, its sub-elements, and the citation that identifies its quotation. This section records how that table was built, the choices made along the way, and the basis for each. Two sources are joined. K&K's Index A and Index B, read from a scan of the 1952 monograph, give each definition its author, year and sub-elements. Boroch's (2016) article gives each definition a quotation with page numbers. Every join follows a stated rule, and each judgment is recorded with its basis in the project's provenance log.

A.1 Sources and the unit

The unit of analysis is the definition K&K attached to each definition, which corresponds to one printed index entry in K&K, and one quotation in Boroch (2016). Boroch's appendix has 171 definitions. Boroch's Table 2 has 135 columns, 132 of which match an appendix definition. The two sources number definitions differently. Boroch renumbers within each group, so his A10 (Winston, 1933) is K&K's A-9. Definitions are therefore matched by author, year, group and subgroup. Positional agreement is kept only as a check. Of 100 matches made by name and year, 48 agree with the position.

A.2 Index A and Index B

Index A (authors) and Index B (sub-elements) were read from the OCR text layer and corrected by explicit rules. The corrections are recorded in a manual-fix table of 142 rows, each with its printed-page evidence. A further 10 rows recover definitions that the OCR glued into element names, and 14 rows record name decisions. The project avoids general character rules, because a rule such as reading a y as a 5 silently produced definitions from unrelated tokens. Each manual fix is keyed to a parsed token, and the checks confirm that every key matches one.

The universe is 171 definitions. Of these, 169 appear in K&K's author index. The other two, C-I-18a (Maquet 1949) and D-II-16 (Aberle et al. 1950), are cited in main-text footnotes but not listed in the index; their Index B entries are read from the sub-element entries that carry them. Four Part III statement definitions (C-5, D-5, D-7 and E-17) are excluded, since they are K&K's statements rather than Part II definitions.

A.3 Matching Boroch's definitions to K&K's

The crosswalk matches each column of Boroch's Table 2 to a K&K definition. Of the 135 columns, 117 match automatically when exactly one definition fits, and 15 more are settled by hand decisions, each with a basis. Three of Boroch's Table 2 columns have no appendix entry of their own and carry no K&K definition. A10a, C1-5a and F4-6. Each is a verbatim copy of another quotation, which leaves 132 columns with a K&K definition. A10a repeats A10 (Winston, 1933), C1-5a repeats C1-5 (Firth, 1939), and F4-6 repeats F4-5, which is the entry Boroch's numbering misprints as F4-6. None of the three adds a quotation, so the object does not use them. The F4-5 quotation is Kluckhohn and Kelly 1945 (printed p. 87), and it gives K&K's F-IV-5. K&K's printed definitions A-10a (Lowie, 1937) and C-I-5a (Lynd, 1940) take their quotations from the appendix entries A12 and C1-6.

Table 6: The four split definitions and their variants.

K&K definition	Variant	Boroch definition and source	Page numbers
A-3	A-3(a)	A3, Dixon 1928	3
A-3	A-3(b)	A4, Dixon 1928	3
E-5	E-5(a)	E5, Linton 1945 (footnote 5)	5
E-5	E-5(b)	E6, Linton 1945	32
F-II-4	F-II-4(a)	F2-4, Blumenthal 1937 (footnote 4)	3
F-II-4	F-II-4(b)	F2-5, Blumenthal 1937	12
F-II-7	F-II-7(a)	F2-8, Feibleman 1946 (tentative; footnote 7)	75
F-II-7	F-II-7(b)	F2-9, Feibleman 1946 (final)	76

A.4 Quotations and citation keys

Each definition has one quotation. Boroch supplies 170 of them; the 132 definitions from Boroch's Table 2 and the 39 appendix-only definitions, less the Sears addendum (C1-22), which is linked to a specialty definition rather than to a network definition. D-III-4 has no Boroch entry, and its quotation is K&K's printed footnote 14 (Roberts 1951, pp. 3 and 6, printed p. 60). Four quotations are not in English and are rendered from a translation table. B-17 and C-I-18a from French, and F-I-5 and F-II-3 from German. C-I-18a uses K&K's own printed English. The other three are working translations that have not been checked on the page images.

Each definition is identified by a citation key. Boroch's author, year and page numbers, together with the first six words of the quotation. The key is unique for all 171 definitions, with no missing values. The page numbers are Boroch's as printed and have not been checked against the source works. The key therefore identifies a quotation as Boroch reports it, and the page numbers are the first thing to check against a source.

A.5 Splitting the a and b quotations

Boroch quotes parts of four K&K definitions separately, labeled (a) and (b). K&K prints one index entry for each of these definitions, so the object had one definition where Boroch has two quotations. On 2026-10-30 the project split each of the four definitions into two variants (Table 6). The split follows Boroch's quotations and is not a distinction K&K prints. Each variant takes its own quotation and its own citation key.

Because K&K gives one index entry per definition, each variant starts with a copy of its parent's sub-element links, 41 in all. A lexical screen then assigns 29 of the 41 to one variant. A link stays on the variant whose quotation contains the sub-element's name and not on the other. The 12 links where both or neither quotation contains the name remain on both variants. The screen is a word match, so a name used in a generic sense, such as "total" or "past," can be assigned wrongly. Each assignment is a reading of the quotation text and should be treated as a judgment. Removing a link also removes the heading edge that carries it, which accounts for 34 edge rows. Together the split takes the verified network from 167 to 171 definitions and from 1189 to 1201 edges.

A.6 Checks, limitations and open items

The networks depend on the K&K Index B transcription and on the verified-definition filter. Element names are not unique across headings, so same-named elements are merged, which can join unrelated entries. Several readings depend on OCR corrections recorded in the repository's

manual-fix files. Two definitions that are not in Index A are included. C-I-18a and D-II-16, from the main text. K&K's Part III statements (C-5, D-5, D-7, E-17) are excluded. The basis for each is documented in the repository. The four split definitions and the 29 link decisions that separate their variants (see the data section) are judgments; the variants share one K&K index entry, and the text-based assignment of concept links is a reading that a reader can check against the quotations, not a printed fact. The definition-level trees and the topics depend on these decisions. Each projection has many maximum spanning trees, because its edge weights tie heavily. Betweenness averages over 1000 random trees, but the trees in the figures show one representative member, so tree-level structure (neighbors, communities and regions) describes that member, and the ranks below the first place in Table 3 vary across trees.

The build's validators check the network's structure, its round trips and its attributes, but not the content of each edge. The block check places 1301 of the 1326 Index B edge rows (before the 34-row removal) in their printed block, and 25 in another block. The page-image review covers 23 of those 25. Several items remain open. The year of K&K's footnote 5, which Boroch gives as Linton 1945a, was read from the page image at 150 dpi, and a higher-resolution check is still needed. The three duplicate columns need recorded decisions in the crosswalk, and the text analysis must be brought into line with them. The three non-English translations need checking against the page images. The Boroch page numbers need checking against the source works. The 29 link assignments are judgments on the text, as described above.

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